



WELCOME TO THE THUNDERDOME

OneFlip: A Handy Spice Jar

Final Project Presentation

Harry Ye Brian Khuu Yunjong Na

ENTR 390 - 3D Printing

Scenario

Imagine you're cooking dinner. You just added beef and veggies into the frying pan, holding spatula on one hand and controlling the pan the other.

Now you realized you need to add some secret seasoning immediately, before the meat is burnt to be too stiff – What would happen?



Problem

Most ingredient containers require two hands to operate, causing unnecessary hassle when the cook is working on the items on the stove.

Reach → Hold → Open → Pour

!! Two Hands All the Time !!

Our design streamlined "holding" and "opening" step, with a single-hand interaction – So the other hand is free to control the pan (e.g. stirring)



Fact

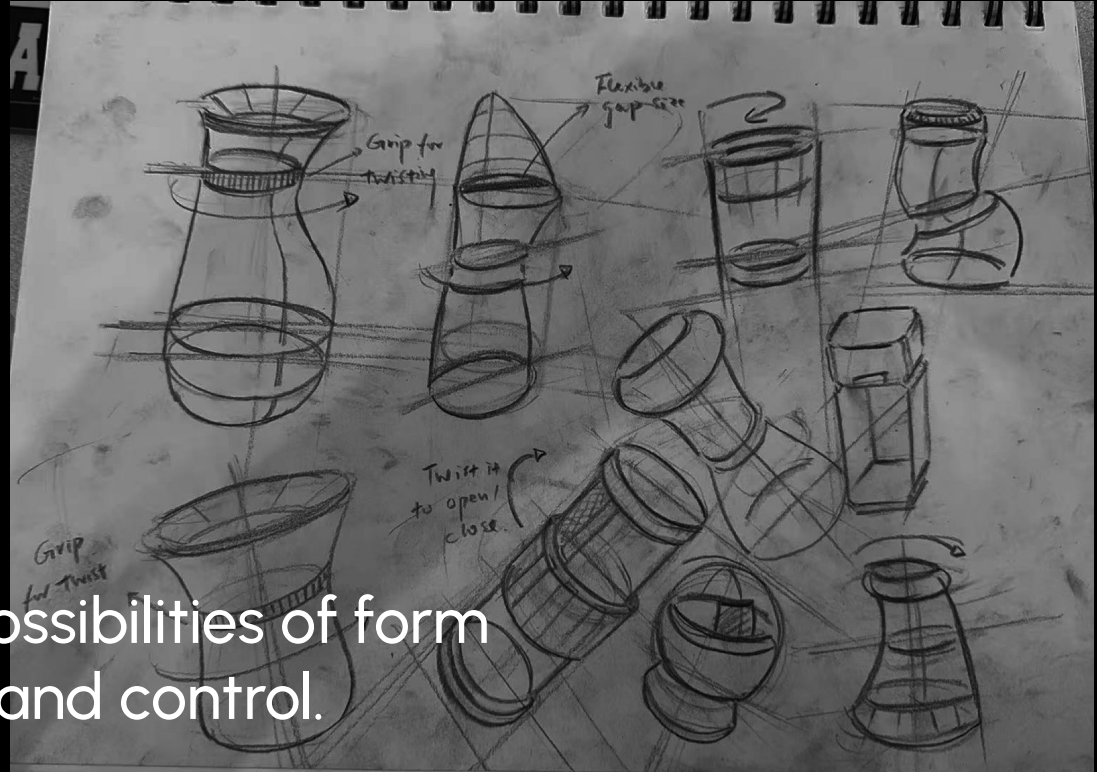
We conducted a desk research on the usage of daily home appliances, targeting the elder specifically.

One study of **750 adults** aged 20–94 found that among people over 50, **only about 48% had no problems opening jars**; around 16% had great difficulty and almost 4% couldn't open a jar at all without help. Our OneFlip concept is trying to live in that 'easy-to-open for almost everyone' zone, but using grip, leverage, and one-hand design instead of just brute force."



Ideation

We ideated various possibilities of form that supports single-hand control.

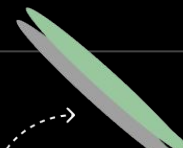


Design



OneFlip reliably supports single hand interaction with excellent ergonomics.

Press



OneFlip reliably supports single hand interaction with excellent ergonomics.

1. **Grip Region:** Create more stable contact with 4 fingers while holding
2. **Sloped Back:** A slight convex inward, suggesting the affordance for the thumb to rest
3. **Well-Positioned Trigger:** When holding correctly, flipping triggers falls nicely within the range
4. **Force-Saving Lever:** We extended the pressing region for the twistable cap, making sure it's triggered with minimal pressure



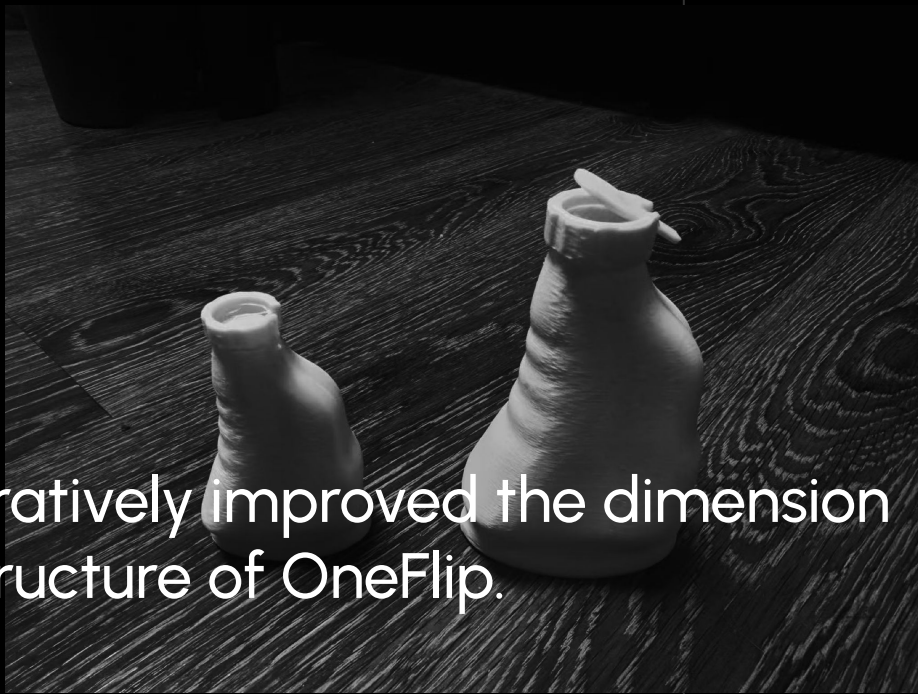
Appearance

OneFlip is designed to be minimal yet functionally reliable.

1. **Organic form :** Curved, smooth surface minimizes harm while grabbing the container without full attention
2. **Transparency:** Ensure the amount of spices remaining is clearly visible
3. **Heavier Bottom:** Stacked in kitchen, spices can spill over easily by accident. We increased contact area with supporting surface to ensure stability.
4. **Components:** Consists of 3 components, easy to assemble and cheap to produce



Before & After



Iteration

We iteratively improved the dimension and structure of OneFlip.

OneFlip 2.0

Precisely fit hand size (12cm)
Shelled as container (0.5mm)
Stabilized cap structure.





Have a great meal!